1. $$SSE(b_0, b_1, b_2) = \sum_{i=1}^{N} (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})^2$$

2. $$\frac{\partial}{\partial b_0} = -2\sum_{i=0}^{N}(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = -2\sum_{i=0}^{N} e_i$$

$$\frac{\partial}{\partial b_1} = -2\sum_{i=0}^{N} z_{i1}(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = -2\sum_{i=0}^{N} z_{i1} e_i$$

$$\frac{\partial}{\partial b_2} = -2\sum_{i=0}^{N} z_{i2}(y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) = -2\sum_{i=0}^{N} z_{i2} e_i$$

3. $\sum_{i=1}^{N} e_i = 0$ avg. error $= \frac{1}{N} \cdot 0 = 0$

$\sum_{i=1}^{N} z_{i1} e_i = 0$ $\sum_{i=1}^{N} z_{i2} e_i = 0$ so $e \cdot z_{i1} = 0$
$e \cdot z_{i2} = 0$.

4. $$\sum_{i=1}^{N} y_i - Nb_0 - b_1\sum_{i=1}^{N} z_{i1} - b_2 \sum_{i=1}^{N} z_{i2} = 0$$

$\rightarrow \sum_{i=1}^{N} y_i - Nb_0 = 0$ $b_0^* = \frac{1}{N}\sum_{i=1}^{N} y_i = \bar{y}$

$$\sum_{i=1}^{N} z_{i1}(y_i - \bar{y}) - b_1 \sum_{i=1}^{N} z_{i1}^2 - b_2 \sum_{i=1}^{N} z_{i1} z_{i2} = 0$$

$$b_1 \sum_{i=1}^{N} z_{i1}^2 + b_2 \sum_{i=1}^{N} z_{i1} z_{i2} = \sum_{i=1}^{N} z_{i1}(y_i - \bar{y})$$

$$\sum_{i=1}^{N} z_{i2}(y_i - \bar{y}) - b_1 \sum_{i=1}^{N} z_{i1} z_{i2} - b_2 \sum_{i=1}^{N} z_{i2}^2 = 0$$

$$b_1 \sum_{i=1}^{N} z_{i1} z_{i2} + b_2 \sum_{i=1}^{N} z_{i2}^2 = \sum_{i=1}^{N} z_{i2}(y_i - \bar{y})$$

$\sum z_{ij} = 0$ so $\sum_{i=1}^{N} z_{ij}(y_i - \bar{y}) = \sum_{i=1}^{N} z_{ij} y_i$

5. $$\begin{bmatrix} \sum z_{i1}^2 & \sum z_{i1} z_{i2} \\ \sum z_{i1} z_{i2} & \sum z_{i2}^2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \sum z_{i1} y_i \\ \sum z_{i2} y_i \end{bmatrix}$$